

LAB 4: CONTROLLING SYSTEM CLOCK AND CLOCK DIVIDING

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Abstract

The purpose of this lab was to use the oscillator feature in assembly using MPLAB IDE to turn LED4 on and off every 0.5s. To do this, the delay needs to be edited so that the delay is as close to 0.5 seconds as possible.

Methodology

The first step in creating this assignment was to create a new file in a new folder to code in assembly in the MPLAB IDE. To do this, the steps on Canvas were followed.

The next step was to initialize the different parts of the program including the libraries, variables and parts of the microcontroller needed during this lab.

The lab's third step was to set up the delays. This was done by defining D1, D2 and D3. These delays were then set to the values of 250, 249, and 4. These values were found by calculating the number of loops needed. Once the delay is called, the function can run. The function works by looping D1 from 250 to 0. After D1 finishes, D2 loops. D2 loops from 249 to 0 and then sends the loop back to D1 to cycle again. Once D2 has completed its cycles, the delay then moves to loop 3. This tells D2 to cycle 4 times. This calculation rounds to 0.500001s.

Once the program is built and downloaded, the program ran as expected. This means that LED4 flipped between on and off every 0.5 seconds.

Project Requirements

The first requirement for this project was to create a new file in a new folder to code in assembly in the MPLAB IDE. To do this, the steps on Canvas were followed.

The next step was to initialize/setting up the different parts of the circuits including: libraries, OSCCON, PORTB, TRISB, etc.

The third requirement for this program was to configure the board oscillator at 8Mhz and to edit the delay function so that LED4 turns on and off every 0.5s.

Design Diagram

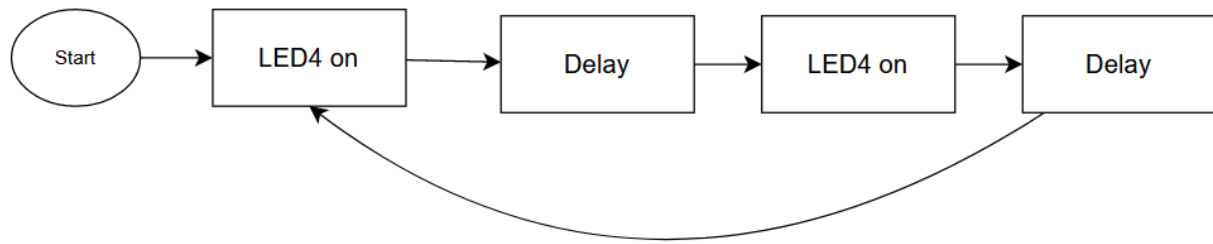


Figure 1: Design Diagram

Code

```
#include <p16F886.inc>
```

```
__CONFIG _CONFIG1, _FOSC_INTRC_NOCLKOUT & _LVP_OFF & _FCMEN_OFF & _IESO_OFF &  
_BOR_OFF & _CPD_OFF & _CP_OFF & _MCLRE_OFF & _PWRTE_ON & _WDT_OFF &  
_INTRC_OSC_NOCLKOUT
```

```
__CONFIG _CONFIG2, _WRT_OFF & _BOR21V
```

```
cblock 0x70
```

```
D1
```

```
D2
```

```
D3
```

```
endc
```

```
ORG 0x0000
```

```
;goto StartLoop
```

```
StartLoop:
```

```
    banksel ANSELH    ;make the digital mod
```

```
    CLRF ANSELH;
```

```
    MOVLW B'00000000';
```

```
    MOVWF ANSELH
```

```

banksel PORTB

CLRF PORTB

banksel TRISB

MOVLW B'11110000'

MOVWF TRISB

banksel OSCCON ;INTCON select

;bsf OSCCON, 7

bsf OSCCON, 6

bsf OSCCON, 5

bsf OSCCON, 4

;bsf OSCCON, 3

;bsf OSCCON, 2

;bsf OSCCON, 1

;bcf OSCCON, 0

;banksel IOCB

;bsf IOCB, 5

movlw 0xFA

movwf D1

movlw 0xF9

movwf D2

movlw 0x4

movwf D3

call Delay

OFFLoop:

banksel PORTB ; select bank

bcf PORTB,0 ; turn OFF LED B3 (DS4)

call Delay ; delay of 1 second

```

goto ONLoop ; go to on loop

ONLoop:

banksel PORTB

bsf PORTB,0 ; turn ON LED B3 (DS4)

call Delay ; delay of 1 second

goto OFFLoop ; go to off loop

Delay:

nop

decfsz D1, 0x1

goto Delay

movlw 0xFA

movwf D1

decfsz D2, 0x1

goto Delay

movlw 0xF9

movwf D2

decfsz D3, 0x1

goto Delay

movlw 0x4

movwf D3

return

end

Results

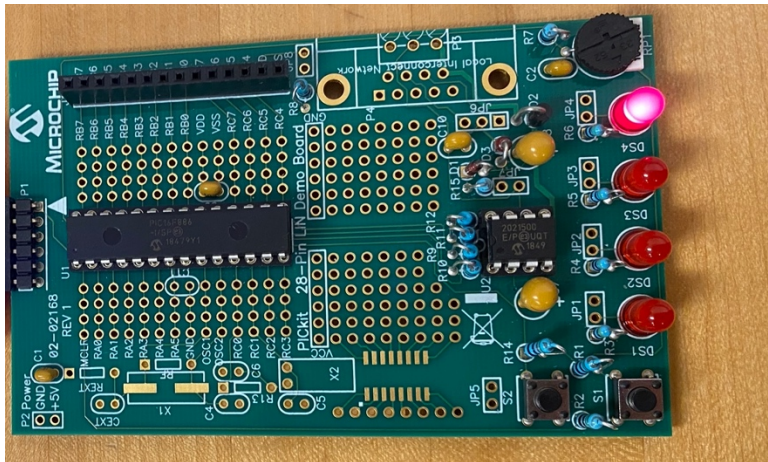


Figure 2: LED4 Flashing On

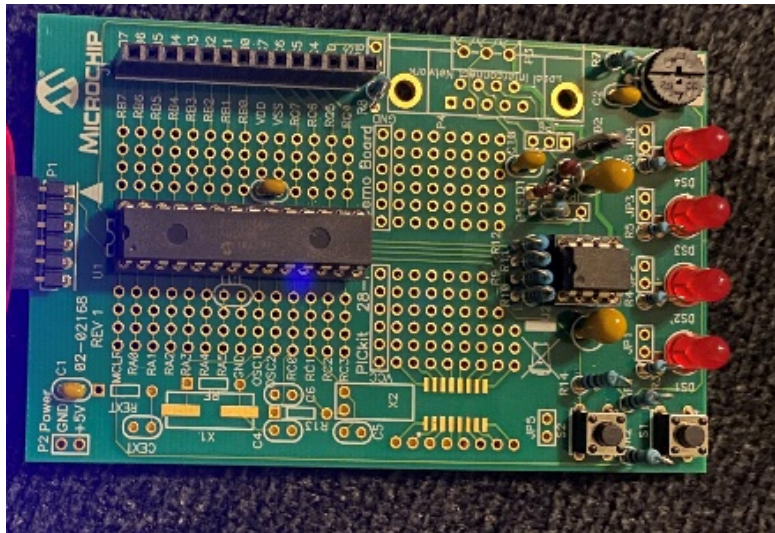


Figure 3: LED4 Flashing Off

Conclusion

In conclusion, the lab was successful. The delay was used by using integrated loops that cycle 250 bits over and over again. These loops, along with calculations, allowed the delay to be almost exactly 0.5s. This meant that LED4 flashed on and off for 0.5 seconds each.